EGH413 Advanced Dynamics Equation Sheet

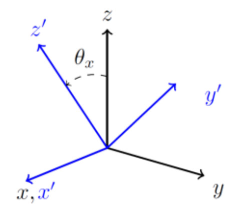
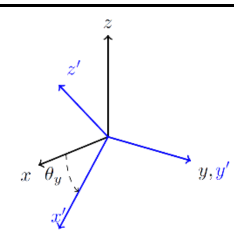
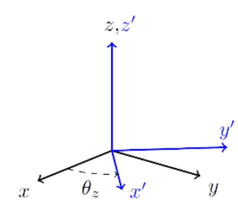
General

$\bar{v}_P = \dfrac{dr_{P/O}}{dt}$	$\bar{a}_P = \dfrac{d\bar{v}_P}{dt}$	$\dot{\bar{A}} = \dfrac{\partial A}{\partial t} + \bar{\omega} \times \bar{A}$
$T_2 + V_2 = T_1 + V_1 + W_{1\to 2}^{nc}$	$W_{1\to 2} = \oint_1^2 \sum \bar{F}(\bar{r}) \cdot d\bar{r}$	$H_O = \sum(\bar{r}_i \times m_i \bar{v}_i)$

Position, velocity, and acceleration of a point P

System	$\boldsymbol{\bar{r}_{P/O}}$	$\boldsymbol{\bar{v}_P}$	$\boldsymbol{\bar{a}_P}$
Cartesian	$x\hat{\imath} + y\hat{\jmath} + z\hat{k}$	$\dot{x}\hat{\imath} + \dot{y}\hat{\jmath} + \dot{z}\hat{k}$	$\ddot{x}\hat{\imath} + \ddot{y}\hat{\jmath} + \ddot{z}\hat{k}$
Cylindrical	$R\bar{e}_R + z\bar{e}_z$	$\dot{R}\bar{e}_R + R\dot{\theta}\bar{e}_\theta + \dot{z}\bar{e}_z$	$(\ddot{R} - R\dot{\theta}^2)\bar{e}_R + (R\ddot{\theta} + 2\dot{R}\dot{\theta})\bar{e}_\theta + \ddot{z}\bar{e}_z$
Spherical	$r\bar{e}_r$	$\dot{r}\bar{e}_r + r\dot{\theta}\sin\phi\ \ \bar{e}_\theta$ $+ r\dot{\phi}\bar{e}_\phi$	$[\ddot{r} - r\dot{\phi}^2 - r\dot{\theta}^2\sin^2\phi]\bar{e}_r$ $+[r\ddot{\phi} + 2\dot{r}\dot{\phi} - r\dot{\theta}^2\sin\phi\cos\phi]\bar{e}_\phi$ $+[r\ddot{\theta}\sin\phi + 2\dot{r}\dot{\theta}\sin\phi + 2r\dot{\phi}\dot{\theta}\cos\phi]\bar{e}_\theta$

Rotation Matrices

Rotation Type	**Rotation Matrix**	**Visualization**
x-axis	$[R_x] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_x & \sin\theta_x \\ 0 & -\sin\theta_x & \cos\theta_x \end{bmatrix}$	
y-axis	$[R_y] = \begin{bmatrix} \cos\theta_y & 0 & -\sin\theta_y \\ 0 & 1 & 0 \\ \sin\theta_y & 0 & \cos\theta_y \end{bmatrix}$	
z-axis	$[R_z] = \begin{bmatrix} \cos\theta_z & \sin\theta_z & 0 \\ -\sin\theta_z & \cos\theta_z & 0 \\ 0 & 0 & 1 \end{bmatrix}$	
Sequence of Body Fixed Rotations	$[R] = [R_n] \ldots [R_2][R_1]$	
Sequence of Space Fixed Rotations	$[R] = [R_1] \ldots [R_{n-1}][R_n]$	